Today

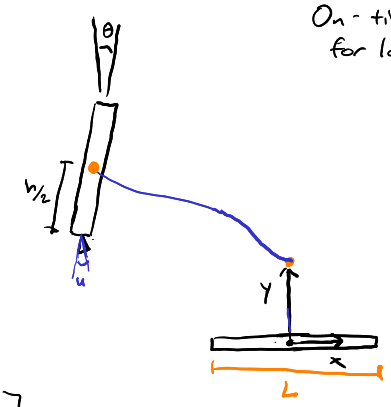
Optimization

Announcements

HW 5 due tomorrow

On-time attendance for lab





$$\vec{x} = \begin{bmatrix} x \\ y \\ \theta \\ \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix} \qquad \dot{\vec{x}} = f(t, \vec{x}, u)$$

↑ dynamics

$$\text{minimize } w_1 \dot{x}(T)^2 + w_2 \dot{y}(T)^2 + w_3 \left(y(T) - \frac{h}{2}\right)^2 + w_4 x(T)^2 + w_5 \int_0^T u(t)^2 \, dt$$

$$\text{subject to } -\frac{L}{2} \le x(T) \le \frac{L}{2}$$

$$u_{min} \le u \le u_{max}$$

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1) Formulation
2) Solution

(1) ↔ (2): often related

Formulation often the hardest

Solution often with an off-the shelf solver fmincon

Unconstrained Optimization Problem

$$\underset{\vec{x}}{\text{minimize}} \; f(\vec{x})$$

sense: minimize

objective (must return a scalar): $f(\vec{x})$

$\vec{x}$: decision variables (can be scalar or vector)

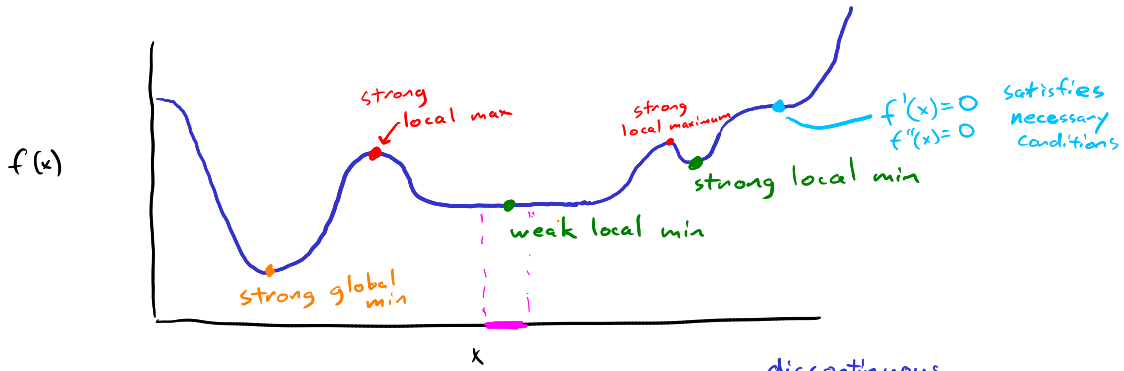
Solutions

- x^* is a global min. if $f(x^*) \le f(x) \quad \forall x$ (both sides)
- x^* is a local min. if $f(x^*) \le f(x) \quad \forall x$ in a neighborhood of x^*

x^* is a strong local/global minimum if $f(x^*) < f(x) \quad \forall x \neq x^*$ in a neighborhood of x^*

x^* is a weak " " minimum if it is a minimum that is not strong

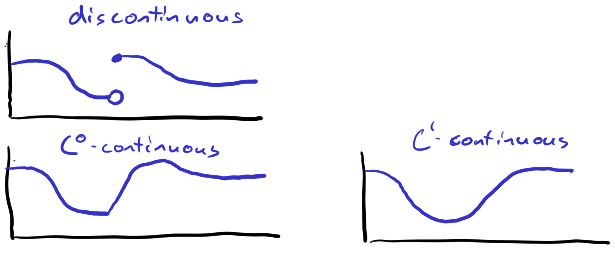
"extremum" can refer to either a maximum or minimum



Smoothness

C^0 continuous: f is continuous

C^1 continuous: f' is continuous



Local Optimality Conditions for C^1 functions

Minimum:

$f'(x^*) = 0$ F.O.N.C. first order necessary condition

"stationary points" satisfy this

$f''(x^*) \geq 0$ S.O.N.C.

necessary, but not sufficient

Example of sufficient conditions:

$f'(x^*) = 0$, $f''(x^*) > 0$ ← not true of weak local minimum

x is a scalar

Local optimality condition for C^1 functions of vectors

F.O.N.C. $\nabla f(\vec{x}^*) = \vec{0}$ $\nabla f(\vec{x}) = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \vdots \\ \frac{\partial f}{\partial x_n} \end{bmatrix}$

S.O.N.C. $H(\vec{x}^*)$ is positive semi-definite

"Hessian" $H(\vec{x}) = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \frac{\partial^2 f}{\partial x_1 \partial x_2} & & \\ \vdots & & \\ \frac{\partial^2 f}{\partial x_1 \partial x_n} & & \frac{\partial^2 f}{\partial x_n^2} \end{bmatrix}$